

Assignment-7.3

2303A51242

Batch-05:

Task-01: Fixing Syntax Errors – Prompts (Detect the error)

The following Python function has a syntax error. Identify the error and explain why it occurs.

```
assignment-7.py > ...  
1 def add(a, b)  
2     return a + b  
3
```

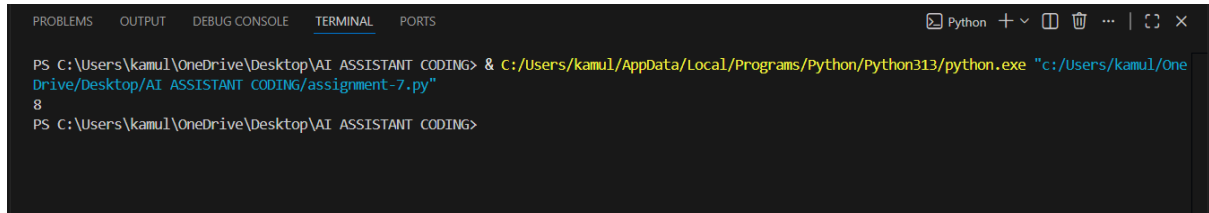
Output:

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [X]  
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Use  
Drive/Desktop/AI ASSISTANT CODING/assignment-7.py"  
File "c:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING\assignment-7.py", line 1  
    def add(a, b)  
            ^  
SyntaxError: expected ':'  
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

SyntaxError- missing : after function definition.

```
assignment-7.py  
assignment-7.py  
assignment-7.py > ...  
1 def add(a, b):  
2     return a + b  
3  
4 print(add(5, 3))  
5
```

Output:

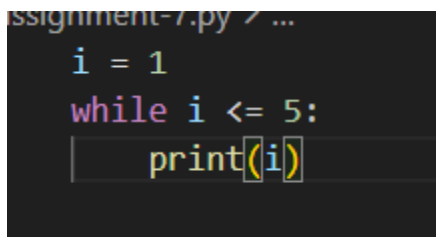


```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\kamu1\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamu1/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamu1/OneDrive/Desktop/AI ASSISTANT CODING/assignment-7.py"
8
PS C:\Users\kamu1\OneDrive\Desktop\AI ASSISTANT CODING>
```

Explanation:

- Python requires a colon (:) after function definitions.
- Without it, the interpreter cannot identify the function body.
- Adding the colon resolves the syntax error and allows execution.

Task-02: Debugging Logic Errors in Loops



```
assignment-7.py / ...
i = 1
while i <= 5:
    print(i)
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
1
1
1
1
1
1
1
1
1
1
1
1
```

- The variable `i` is never incremented.
- Condition always remains true → infinite loop.

```
assignment-1.py  assignment-5.py  assignment-7.py > ...
1   i = 1
2   while i <= 5:
3       print(i)
4       i += 1
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + - [ ] [ ] [ ]

PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-7.py"
1
2
3
4
5
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Explanation:

- Loop variables must change inside the loop.
- Incrementing i ensures loop termination.
- AI identified missing increment as the logic error.

Task-03: Handling Runtime Errors (Division by Zero)

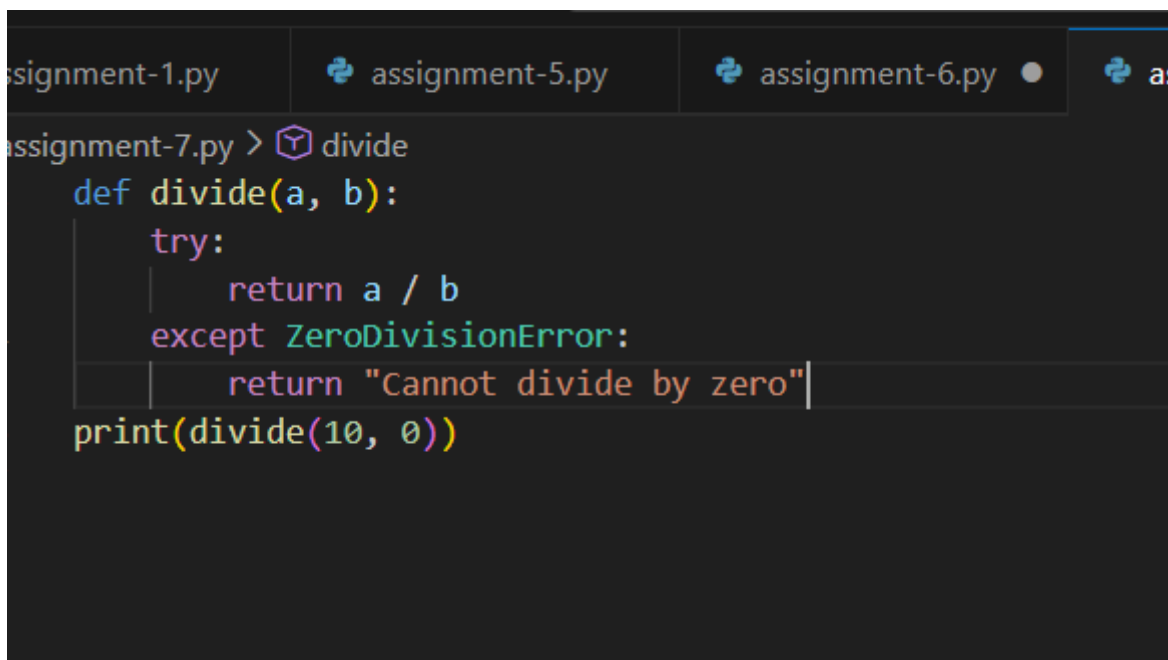
```
assignment-7.py > ...
1   def divide(a, b):
2       return a / b
3
4   print(divide(10, 0))
```

Output:



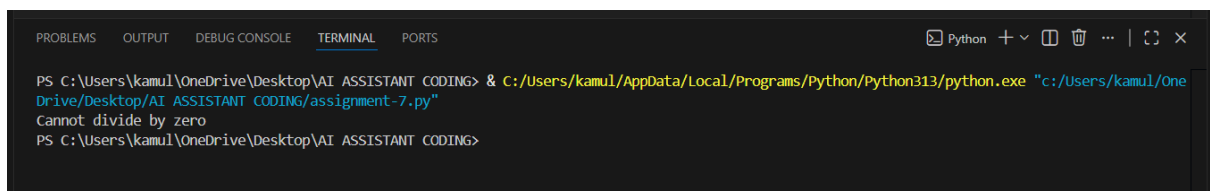
```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-7.py"
Traceback (most recent call last):
  File "c:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING\assignment-7.py", line 4, in <module>
    print(divide(10, 0))
    ~~~~~^~~~~~
  File "c:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING\assignment-7.py", line 2, in divide
    return a / b
    ~~~~~^~~~~
ZeroDivisionError: division by zero
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Runtime Error: ZeroDivisionError.



```
assignment-7.py > def divide(a, b):
    try:
        return a / b
    except ZeroDivisionError:
        return "Cannot divide by zero"
print(divide(10, 0))
```

Output:



```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-7.py"
Cannot divide by zero
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Explanation:

- Division by zero causes program crash.
- try-except prevents abnormal termination.
- AI suggested graceful error handling for runtime safety.

Task-04: Debugging Class Definition Errors

```
assignment-7.py > ...
1 class Student:
2     def __init__(name, roll):
3         name = name
4         roll = roll
5
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [X] ... | [ ] [X]
```

```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-7.py"
```

```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

- self parameter is missing.
- Instance variables are not properly assigned.

```
assignment-1.py assignment-5.py assignment-6.py assignment-7.py X
assignment-7.py > ...
1 class Student:
2     def __init__(self, name, roll):
3         self.name = name
4         self.roll = roll
5
6 s = Student("Amulya", 101)
7 print(s.name, s.roll)
8
```

Output:



The screenshot shows a Windows command prompt window with the title bar "Python". The window has tabs for "PROBLEMS", "OUTPUT", "DEBUG CONSOLE", "TERMINAL" (which is active), and "PORTS". The terminal shows the following commands and output:

```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:\Users\kamul\AppData\Local\Programs\Python\Python313\python.exe "c:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING\assignment-7.py"
Amulya 101
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Explanation:

- self refers to the current object.
- Without self, Python cannot store instance data.
- AI fixed constructor and attribute assignment.

Task-05: Resolving Index Errors in Lists

```
assignment-7.py > ...  
1 numbers = [10, 20, 30]  
2 print(numbers[5])
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python + ~ [Icons]
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:\Users\kamul\AppData\Local\Programs\Python\Python313\python.exe "c:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING\assignment-7.py"
Traceback (most recent call last):
  File "c:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING\assignment-7.py", line 2, in <module>
    print(numbers[5])
            ~~~~~^~
IndexError: list index out of range
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

IndexError: list index out of range.

```
assignment-7.py > ...
1
2 numbers = [10, 20, 30]
3 index = 2
4
5 if index < len(numbers):
6     print(numbers[index])
7 else:
8     print("Index out of range")
9 try:
10     print(numbers[5])
11 except IndexError:
12     print("Index out of range")
```

Output:



The screenshot shows a VS Code terminal window with the following tabs: PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The terminal displays the following text:

```
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING> & C:/Users/kamul/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/kamul/OneDrive/Desktop/AI ASSISTANT CODING/assignment-7.py"
30
Index out of range
PS C:\Users\kamul\OneDrive\Desktop\AI ASSISTANT CODING>
```

Explanation:

- Lists have fixed indices from 0 to `len(list)-1`.
- AI suggested bounds checking and exception handling.
- Prevents program crash due to invalid access.